Statistical Test for Means

Why did we use a T-test instead of Z- Test?

The population SD was unknown, so we estimated variability using the sample SD

Formula:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

Xbar = Sample mean
μ0 = Hypothesized population mean
s = Sample SD
n = Sample Size

Degree of freedom = df = n-1

One Sample T-Test

Busines Example: Employee Training Time
An organization claims that employees take an average of 40 minutes to complete a particular task. The manager randomly selects 16 employees to check whther the actual average completion time differs from 40 minutes.

Hypotesized Mean = 40 minutes
Sample Mean = 43 minutes
Sample SD = 6 minutes
Sample Size = 16
Significance level = 0.05

Step 1: Formulate hypothesis
We want to determine whther the mean completion time differs from 40 minutes.

H0:μ=40
H1:μ≠40

This is two tailed test.

Step 2: Calculate SE

SE = s/sqrt(n) = 6/SQRT (16)

SE = 1.5

Step 3: Calculate the t- statistic

t = (43-40)/1.5 = 3/1.5

t = 2

Step 4: Calculate the degree of freedom

df = n-1
df = 16-1

df = 15

Step 5: Find the critical t-value
For two tailed test:

α=0.05,df=15

Using at-distribution table:

$t_{critical} \approx \pm 2.131$

Step 6: Apply the decision rule

For a two tailed test

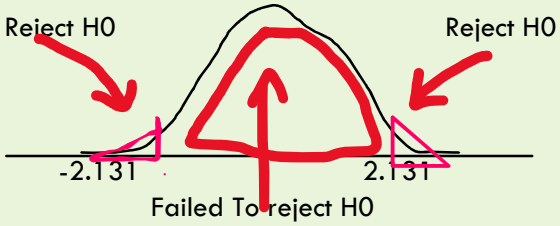
Reject H0 if $|t| > t_{critical}$

Our calculate value is

$|2| < 2.131$

Decision: Fail to Reject H0

This sample does not provide sufficient evidence at 5% significance level that the population mean differs from 40 minutes.



=T.INV.2T(0.05,15)

2.13145

Fail To Reject H0

If the value of t is greater than Tcritical then Reject H0 else Fail to reject H0

Business Interpretation:

The sample average is three minutes above the target, but the difference is not statistically significant at 5% level

This does not prove that the true mean equals 40 minutes. It means the available evidence is not sufficient to reject the null hypothesis.

Independent Two sample T- Test

An indepnedent two sample T-test compares the population means of two independent groups.

Business Example: Comparing two stores

A retail company wants to compare average daily sales at two stores

Parameter	Store A	Store B
Size	10	10
Sample Mean	₹ 520.00	₹ 490.00
Sample SD	₹ 40.00	₹ 35.00

The Population SD is unknown

Step 1: Formulate Hypothesis

H0:μA=μB

H1:μA≠μB

Use two tailed test with Significance level = 0.05

Step 2: Calculate the test statistic

Welch's T-test

$$t = \frac{\bar{X}_A - \bar{X}_B}{\sqrt{\frac{s_A^2}{n_A} + \frac{s_B^2}{n_B}}}$$

Substitute:

t = (520-490)/SQRT(40^2/10 + 35^2/10)

t = 30/SQRT(160+122.5)

t = 1.784892

Df = $\dfrac{\left(\frac{s_A^2}{n_A}+\frac{s_B^2}{n_B}\right)^2}{\frac{\left(\frac{s_A^2}{n_A}\right)^2}{n_A-1}+\frac{\left(\frac{s_B^2}{n_B}\right)^2}{n_B-1}}$

$s^2A/nA = 40^2/10 = 1600/10 = 160$

$s^2B/nB = 35^2/10 = 1225/10 = 122.5$

Numerator

$=(122.5+160)^2 = 282.5^2 =$ 79806.25

Denominator

$=(160^2/10-1)+(122.5^2/10-1)$

$=(25600/9)+15006.25/9$

$=2844.444+1667.361$

$=4511.806$

df = 79806.25/4511.806

17.68832

df = 17.69

df = 17 alpha = 0.05

Tcritical = +-2.110

Step 3 : Compare

|1.785| <2.110

Decision: Fail to Reject

Business Interpretation:

Store A has a higher observed sample mean, but the evidence is insufficient to conclude that the population mean sales differ at the 5% significance level.

Paired Sample T-Test

A paired sample T-test compares two related measurements such as the same employees productivity before and after the training

Business Example: Traing Effectiveness

Employee	before	After	Difference
A	20	25	5
B	24	27	3
C	22	26	4
D	18	22	4
E	26	29	3
		SD =	0.83666

define:

d = after-before

The differences are:

5,3,4,4,3

Step 1: Hypothesis

H0:μd=0

H1:μd≠0

Step 2: Calculate the mean difference:

dbar = (5+3+4+4+3)/5

dbar = 3.8

Step3: Calculate the Standard deviation of differences

Sd = 0.837

Step 4: Calulate t-statistic

$$t = \frac{\bar{d}}{Sd/\sqrt{n}}$$

t = 3.8/0.837/SQRt(5)

t = 10.1518

Degree of freedom:

df = 5-1 = 4

For two tailed test at alpha = 0.05

Tcritical ≈±2.776

Step 5: Make a decision

|10.15| > 2.776

Decision: Reject H0

The data provides statistically significant evidence of a change in the average productivity

Business Interpretation:

All five employees involved in this example. The paired test evaluates whther the average change is statistically distinguishable from zero.

However, this small data does not establish that training alone caused the improvement.

Assumptions of T-Tests

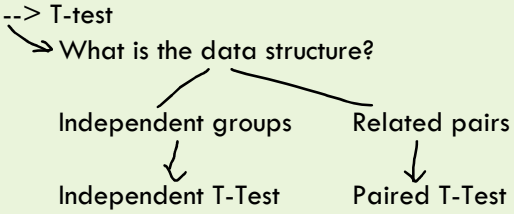
1. Population Standard deviaiton is unknown
2. Random sampling
3. Indepndence
4. Approximate normality
5. No extreme outliers.
6. Equal variances for the pooled two sample t-test

What happen if we have two independent groups with unequal variance/SD?

Use Welch's T-test rather than automatically assuming the equal variance.

Choosing between Z-Test and T-Test

1. What are we testing--> Population mean
2. Is population SD known?
3. YES --> Z-Test , NO --> T-test



Business Problem	Suitable test
Compare a sample mean with a target; population SD known	One-Sample Z- Test
Compare a sample mean with a target; population SD unkknown	One Sample T-Test
Compare means of two independent groups; population SDs Unknown	Independent two-sample T-Test
Compare the same employees before and after the training	Paired Sample t-Test

Decision Rule

for two-tailed test

|t| >tcritical => Reject H0

Otherwise

failed to reject

For a right tailed test

t > Tcritical

For left tailed test:

t < tcritical